

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5 , BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Code of Conduct:

I _____P.Ruthish_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

1. Write pseudocode to simulate a simplified TLS handshake process, including client request, server authentication using a digital certificate, session key generation, and secure data exchange.
2. Develop pseudocode for a firewall rule engine that filters incoming packets based on IP address, port number, and protocol, and decides whether to allow or block traffic.
3. Write pseudocode to detect and classify malware behavior (virus, worm, or Trojan) based on propagation method, file modification patterns, and user interaction.
4. Design pseudocode for an Intrusion Detection System (IDS) that uses both signature-based and anomaly-based detection to identify suspicious network activity.
5. Create pseudocode for an Intrusion Prevention System (IPS) that monitors network traffic in real time and automatically blocks malicious requests while logging intrusion attempts.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

PSEUDOCODE WRITING TASK – QUESTION 4

Title

Design of an Intrusion Detection System Using Signature-Based and Anomaly-Based Detection

1. Aim

To design pseudocode for an **Intrusion Detection System (IDS)** that combines signature-based and anomaly-based detection techniques to identify suspicious network activity and generate appropriate security alerts.

2. Problem Statement

An organization needs an IDS capable of detecting both **known attacks** and **previously unknown or unusual network behavior**.

A signature-based IDS identifies attacks by comparing network activity with known attack patterns, while an anomaly-based IDS identifies activity that significantly deviates from established normal behavior.

The proposed system combines both techniques to improve detection coverage.

3. Objectives

The IDS should:

1. Continuously monitor network traffic.
2. Extract relevant features from each network event.

3. Compare traffic against known attack signatures.
 4. Compare traffic behavior against a normal baseline.
 5. Detect known malicious activity.
 6. Detect abnormal or suspicious activity.
 7. Assign an appropriate severity level.
 8. Generate security alerts.
 9. Log detected events for investigation.
 10. Continue monitoring after processing each event.
-

4. Input

The IDS receives network events containing information such as:

Source IP Address
Destination IP Address
Source Port
Destination Port
Protocol
Packet Size
Connection Rate
Connection Duration
Request Pattern
Payload/Signature Information
Timestamp

5. Output

The IDS produces:

NORMAL
SUSPICIOUS
MALICIOUS

For suspicious or malicious activity, the system generates:

- Alert ID
- Source IP

- Destination IP
 - Detection method
 - Attack/signature information
 - Severity
 - Timestamp
 - Recommended response
-

6. Detection Techniques

6.1 Signature-Based Detection

Signature-based detection compares observed traffic with known patterns of malicious activity.

Network Event

|
v

Extract Features

|
v

Compare with Signatures

|
+--+--+

| |
Match No Match

| |
v v

Malicious Continue

Advantages

- Effective against known attacks.
- Usually produces predictable results.
- Easy to explain when a known signature matches.

Limitation

It may fail to identify completely new attacks for which no signature exists.

7. Anomaly-Based Detection

Anomaly detection compares current network behavior against a baseline of normal behavior.

Example:

Normal connection rate = 20 connections/minute

Observed connection rate = 500 connections/minute

The large deviation may indicate suspicious behavior.

Network Event

|

v

Calculate Behavior

|

v

Compare With Baseline

|

+--+--+

| |

Normal Abnormal

| |

v v

Allow Suspicious

Advantages

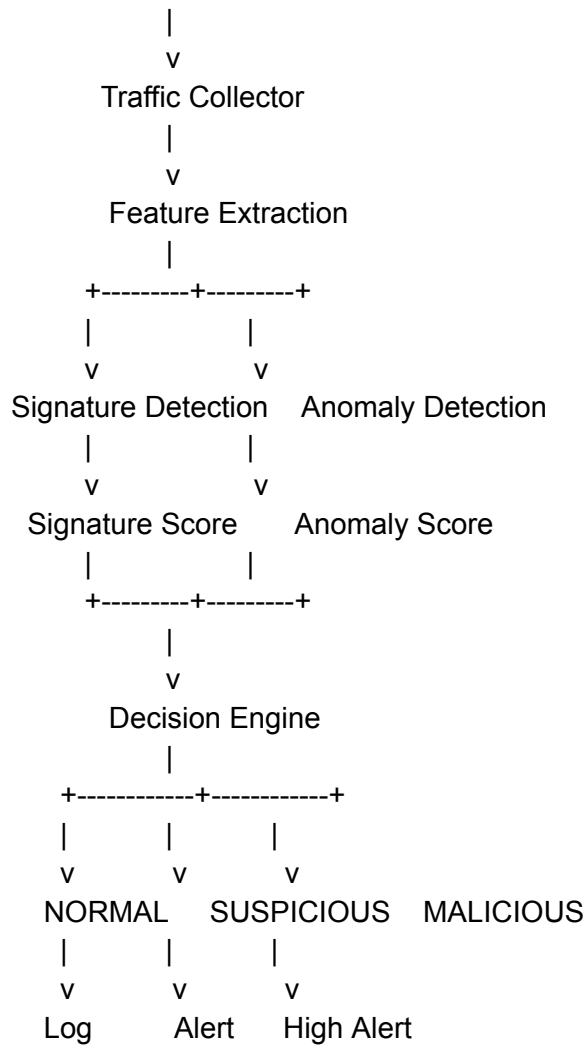
- Can detect previously unknown behavior.
- Useful for detecting unusual activity.
- Can identify attacks that do not have known signatures.

Limitation

It can generate false positives if legitimate behavior differs significantly from the baseline.

8. Combined IDS Architecture

NETWORK TRAFFIC



9. Data Structures

The IDS uses the following logical structures.

Signature Database

Signature_ID
 Attack_Type
 Pattern
 Severity
 Description

Normal Baseline

Protocol
Average Connection Rate
Normal Packet Size
Normal Port Usage
Normal Traffic Volume

Alert Record

Alert_ID
Timestamp
Source_IP
Destination_IP
Detection_Method
Severity
Description

10. Pseudocode

ALGORITHM Hybrid_IDS

INPUT:

Network traffic stream
Signature database
Normal behavior baseline

OUTPUT:

Security alerts and event logs

BEGIN

INITIALIZE signature_database
INITIALIZE normal_baseline
INITIALIZE alert_log

WHILE network monitoring is active DO

 packet ← CAPTURE_NEXT_NETWORK_EVENT()

 IF packet = NULL THEN

```
CONTINUE
END IF
```

```
-----
-- STEP 1: Extract Network Features
-----
```

```
features ← EXTRACT_FEATURES(packet)

source_ip ← features.source_ip
destination_ip ← features.destination_ip
source_port ← features.source_port
destination_port ← features.destination_port
protocol ← features.protocol
packet_size ← features.packet_size
connection_rate ← features.connection_rate
```

```
-----
-- STEP 2: Signature-Based Detection
-----
```

```
signature_match ← FALSE
matched_signature ← NULL
```

```
FOR each signature IN signature_database DO
```

```
    IF MATCH_SIGNATURE(features, signature.pattern) = TRUE THEN
```

```
        signature_match ← TRUE
        matched_signature ← signature
```

```
        BREAK
```

```
    END IF
```

```
END FOR
```

```
-----
-- STEP 3: Anomaly-Based Detection
-----
```

```
anomaly_score ← CALCULATE_ANOMALY_SCORE(  
    features,  
    normal_baseline  
)
```

```
IF anomaly_score >= HIGH_ANOMALY_THRESHOLD THEN
```

```
    anomaly_status ← "HIGH"
```

```
ELSE IF anomaly_score >= LOW_ANOMALY_THRESHOLD THEN
```

```
    anomaly_status ← "MEDIUM"
```

```
ELSE
```

```
    anomaly_status ← "NORMAL"
```

```
END IF
```

```
-----  
-- STEP 4: Decision Engine  
-----
```

```
IF signature_match = TRUE THEN
```

```
    classification ← "MALICIOUS"
```

```
    severity ← matched_signature.severity
```

```
    detection_method ← "SIGNATURE"
```

```
    description ←  
        "Known malicious pattern detected"
```

```
ELSE IF anomaly_status = "HIGH" THEN
```

```
    classification ← "MALICIOUS"
```

```
    severity ← "HIGH"
```

```
    detection_method ← "ANOMALY"
```

```
description ←  
    "Highly abnormal network behavior detected"  
  
ELSE IF anomaly_status = "MEDIUM" THEN  
  
    classification ← "SUSPICIOUS"  
  
    severity ← "MEDIUM"  
  
    detection_method ← "ANOMALY"  
  
    description ←  
        "Network behavior differs from normal baseline"  
  
ELSE  
  
    classification ← "NORMAL"  
  
    severity ← "LOW"  
  
    detection_method ← "NONE"  
  
    description ←  
        "No suspicious activity detected"  
  
END IF
```

```
-----  
-- STEP 5: Logging  
-----
```

```
log_entry ← CREATE_LOG_ENTRY(  
    timestamp,  
    source_ip,  
    destination_ip,  
    classification,  
    detection_method,  
    severity,  
    description  
)
```

```
ADD log_entry TO alert_log
```

```
-----
-- STEP 6: Alert Generation
-----

IF classification = "MALICIOUS" THEN

    GENERATE_ALERT(
        "HIGH PRIORITY",
        log_entry
    )

ELSE IF classification = "SUSPICIOUS" THEN

    GENERATE_ALERT(
        "MEDIUM PRIORITY",
        log_entry
    )

END IF

-----
-- STEP 7: Update Baseline Carefully
-----

IF classification = "NORMAL" THEN

    UPDATE_NORMAL_BASELINE(
        normal_baseline,
        features
    )

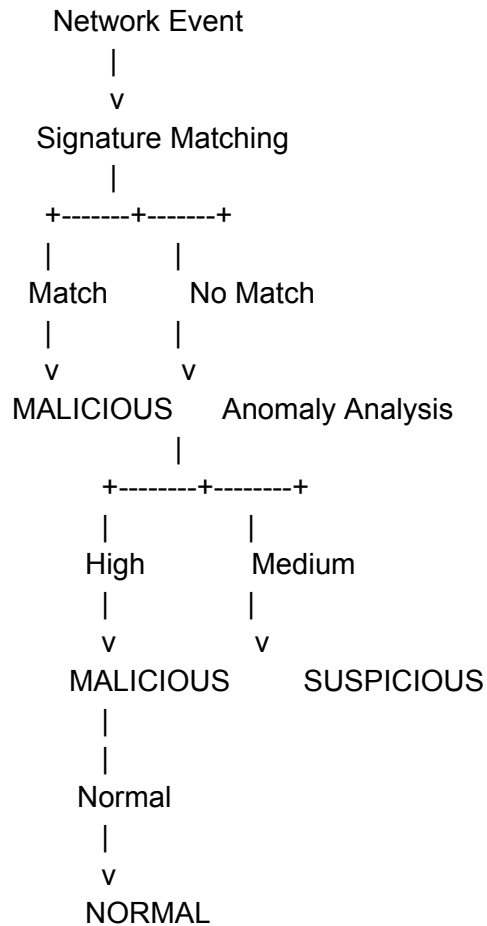
END IF

END WHILE

END
```

11. Detailed Decision Logic

The decision engine follows this priority:



A known malicious signature should take priority because it provides strong evidence of a known attack.

12. Anomaly Score

A simplified anomaly score can be calculated using multiple network characteristics.

For example:

Anomaly Score =
Traffic Rate Deviation
+ Packet Size Deviation
+ Port Usage Deviation

+ Connection Pattern Deviation

The score can then be normalized to a suitable range.

Example:

0 – 30 → Normal

31 – 60 → Suspicious

61 – 100 → Highly Suspicious

The exact thresholds should be established and validated using representative organizational traffic rather than arbitrarily assumed.

13. Example 1 – Known Attack

Suppose an incoming event matches a known attack signature.

Source IP: 10.10.10.25

Destination Port: 445

Signature: Known malicious pattern

Processing:

Packet



Feature Extraction



Signature Database



MATCH



MALICIOUS



High-Priority Alert



Log Event

Result:

Classification: MALICIOUS
Detection Method: SIGNATURE
Severity: HIGH

14. Example 2 – Unknown Abnormal Behavior

Suppose a workstation normally generates:

20 connections/minute

but suddenly generates:

600 connections/minute

No known signature is found.

The anomaly detector observes the significant deviation.

No Signature Match

|

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Anomaly Analysis

|

v

Large Deviation

|

v

High Anomaly Score

|

v

MALICIOUS / HIGH ALERT

This allows the IDS to identify suspicious behavior even when a predefined signature is unavailable.

15. Example 3 – Legitimate Activity

Suppose a backup server normally transfers large amounts of data during a scheduled backup window.

The traffic may appear unusual compared with ordinary user activity.

However, if the behavior is included in the established baseline:

Known Normal Behavior

|

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Baseline Match

|

v

NORMAL

The system avoids generating unnecessary alerts.

This demonstrates why accurate baseline construction is important for reducing false positives.

16. Test Cases

Test Case	Input Condition	Signature	Anomaly	Expected Result
1	Normal web traffic	No match	Low	NORMAL
2	Known malicious pattern	Match	Any	MALICIOUS
3	Extremely abnormal connection rate	No match	High	MALICIOUS
4	Moderately unusual traffic	No match	Medium	SUSPICIOUS
5	Scheduled backup traffic	No match	Low after baseline	NORMAL
6	Normal traffic with unusual port	No match	Medium	SUSPICIOUS

17. Advantages of the Combined Approach

Signature-Based Detection

Provides:

- Accurate detection of known attacks.
- Fast matching.
- Clear explanation of alerts.

Anomaly-Based Detection

Provides:

- Detection of unusual behavior.
- Ability to identify previously unknown attack patterns.
- Detection of deviations from normal network behavior.

Combined Approach

Using both methods provides broader detection coverage.

Known Attacks



Signature Detection

Unknown/Unusual Attacks



Anomaly Detection

Both



Improved IDS Coverage

18. False Positive Management

Anomaly detection can generate false positives.

To reduce them:

1. Build baselines from representative traffic.
 2. Separate users, servers, and network zones where appropriate.
 3. Account for scheduled activities.
 4. Use multiple features rather than a single threshold.
 5. Correlate alerts from multiple sources.
 6. Regularly tune detection thresholds.
 7. Do not automatically classify every anomaly as a confirmed attack.
 8. Continuously evaluate detection performance.
-

19. Complexity Analysis

Assume:

- NN = number of monitored network events.
- SS = number of stored signatures.
- FF = number of extracted features.

A simple sequential signature scan has approximately:

$O(N \times S)$

matching work in the basic model.

Anomaly scoring can be approximately:

$O(N \times F)$

if each event is compared against a fixed-size feature baseline.

In a production system, optimized signature indexes, compiled rules, flow aggregation, parallel processing, and hardware acceleration can improve throughput.

20. Security Benefits

The proposed hybrid IDS provides:

- Detection of known attacks.
 - Detection of abnormal behavior.
 - Continuous network monitoring.
 - Centralized logging.
 - Security alerts.
 - Better visibility into network activity.
 - Support for incident investigation.
 - Improved detection coverage compared with relying on only one technique.
-

21. Limitations

The system also has limitations:

Signature-Based IDS

Cannot reliably detect completely new attacks without an appropriate signature.

Anomaly-Based IDS

May produce false positives when legitimate activity differs significantly from the learned baseline.

Combined IDS

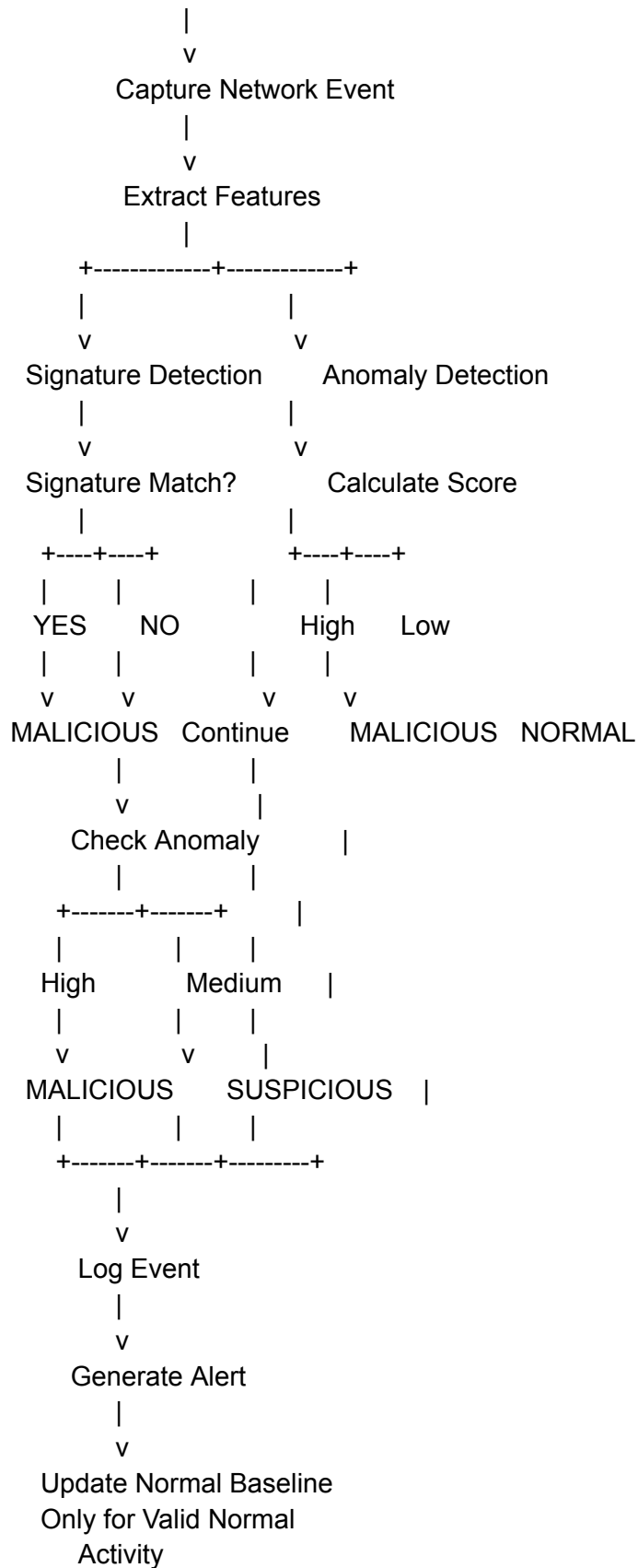
Requires:

- Regular tuning.
- Accurate baseline data.
- Updated signature databases.
- Adequate processing resources.
- Skilled security monitoring.

Therefore, an IDS should be considered a **detection and alerting system**. Blocking traffic automatically is primarily an IPS function.

22. Final Algorithm Flowchart

START



|
v
REPEAT

23. Final Evaluation

Requirement	Proposed Solution
Known attack detection	Signature-based detection
Unknown behavior detection	Anomaly-based detection
Continuous monitoring	Network traffic collector
Alert generation	Decision engine
Event investigation	Centralized logging
False-positive reduction	Baseline tuning and correlation
Detection coverage	Hybrid approach
Automatic blocking	Not performed by IDS; handled by IPS

24. Result

The pseudocode successfully designs a hybrid **Intrusion Detection System** that combines signature-based and anomaly-based detection.

The signature engine identifies known malicious patterns, while the anomaly engine identifies significant deviations from normal network behavior. The decision engine combines both results to classify network activity as **NORMAL, SUSPICIOUS, or MALICIOUS**.

25. Conclusion

A hybrid IDS provides stronger detection coverage than relying exclusively on signature-based or anomaly-based detection.

Signature-based detection is highly useful for identifying known attacks quickly and accurately, while **anomaly-based detection** can identify unusual behavior and potentially detect previously unknown threats.

The proposed algorithm continuously captures network events, extracts features, performs both detection methods, classifies activity, generates alerts, and records events for security analysis.

Therefore, combining both approaches provides a practical foundation for an enterprise IDS, while careful baseline construction, signature updates, threshold tuning, and alert correlation are necessary to maintain reliable detection and minimize false positives.